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Degree: B.Tech Semester: 3rd - Course work

SUMMER SEMESTER (END-SEMESTER) EXAMINATION, JULY, 2024

Course Code: ECECC06/EIECC06

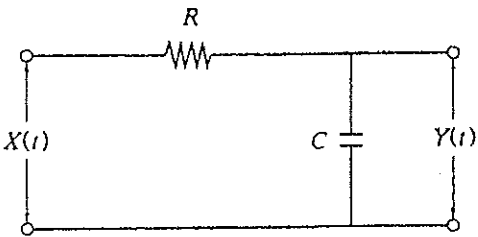
Course Title: Probability Theory and Random Processes

Time: 3 Hours

Max. Marks: 50

Note: Attempt all questions. Missing data/information (if any), may be suitably assumed & mentioned in the answer.

S.No.	Question	Marks	CO
1.	Attempt any two parts		
(a)	Suppose X has the Moment Generating Function: $m_X(t) = (1 - 2t)^{-\frac{1}{2}} \text{ for } t < \frac{1}{2}$ Find the first and second moments of X .	5	CO1
(b)	Calculate mean and variance of binomial distribution.	5	CO1
(c)	Two baskets, basket 1 and basket 2, contain, respectively, two white and one black ball, and one white and five black balls. One ball is transferred from basket 1 to basket 2 and then one ball is drawn from the latter. Suppose the drawn ball happens to be white. What is the probability that the transferred ball was black?	5	CO1
2.	Attempt any two parts		
(a)	Let X and Y be Joint continuous random variables with joint PDF is given below $f_{XY}(x, y) = \begin{cases} 2 & x, y > 0 \text{ and } x + y \leq 1 \\ 0 & \text{otherwise} \end{cases}$ Find co-variance of X and Y	5	CO2
(b)	Explain the following (i) Correlation (ii) Covariance (iii) Correlation coefficient	5	CO2
(c)	Explain marginal probability density function, what is the condition of independent random variables X and Y .	5	CO2
3	Attempt any two parts		
(a)	State and prove the Central Limit Theorem.	5	CO3
(b)	If X is Uniformly distributed in $[-\pi/2, \pi/2]$. Find the p.d.f of $Y = \tan X$.	5	CO3
(c)	State and prove Markov's inequality.	5	CO3

4.	Attempt any two parts		
(a)	The autocorrelation function of a random process is given below. Calculate mean and variance of random process. $R_{XX}(\tau) = 25 + \frac{4}{1+6\tau^2}$	5	CO4
(b)	Define Gaussian Random Process and mention at least four properties	5	CO4
(c)	Define Ergodic Random Process. Given $X(t) = A \cos(\omega t + \theta)$. A and ω are constants and θ is a uniform random variable over $(-\pi, \pi)$. Is $X(t)$ ergodic in both mean and autocorrelation? Justify your answer	5	CO4
5.	Attempt any two parts		
(a)	A WSS random process $X(t)$ with autocorrelation function $R_X(\tau) = e^{-a \tau }$, where a is a real positive constant, is applied to the input of an LTI system with impulse response $h(t) = e^{-bt}u(t)$, where b is a real positive constant. Find $R_Y(\tau)$ for output $Y(t)$.	5	CO5
(b)	The input $X(t)$ to the RC filter shown below is a white noise process.  Determine the power spectrum, auto-correlation and mean-square value of $Y(t)$	5	CO5
(c)	Write down properties of the auto-correlation and cross-correlation function of random process in detail.	5	CO5